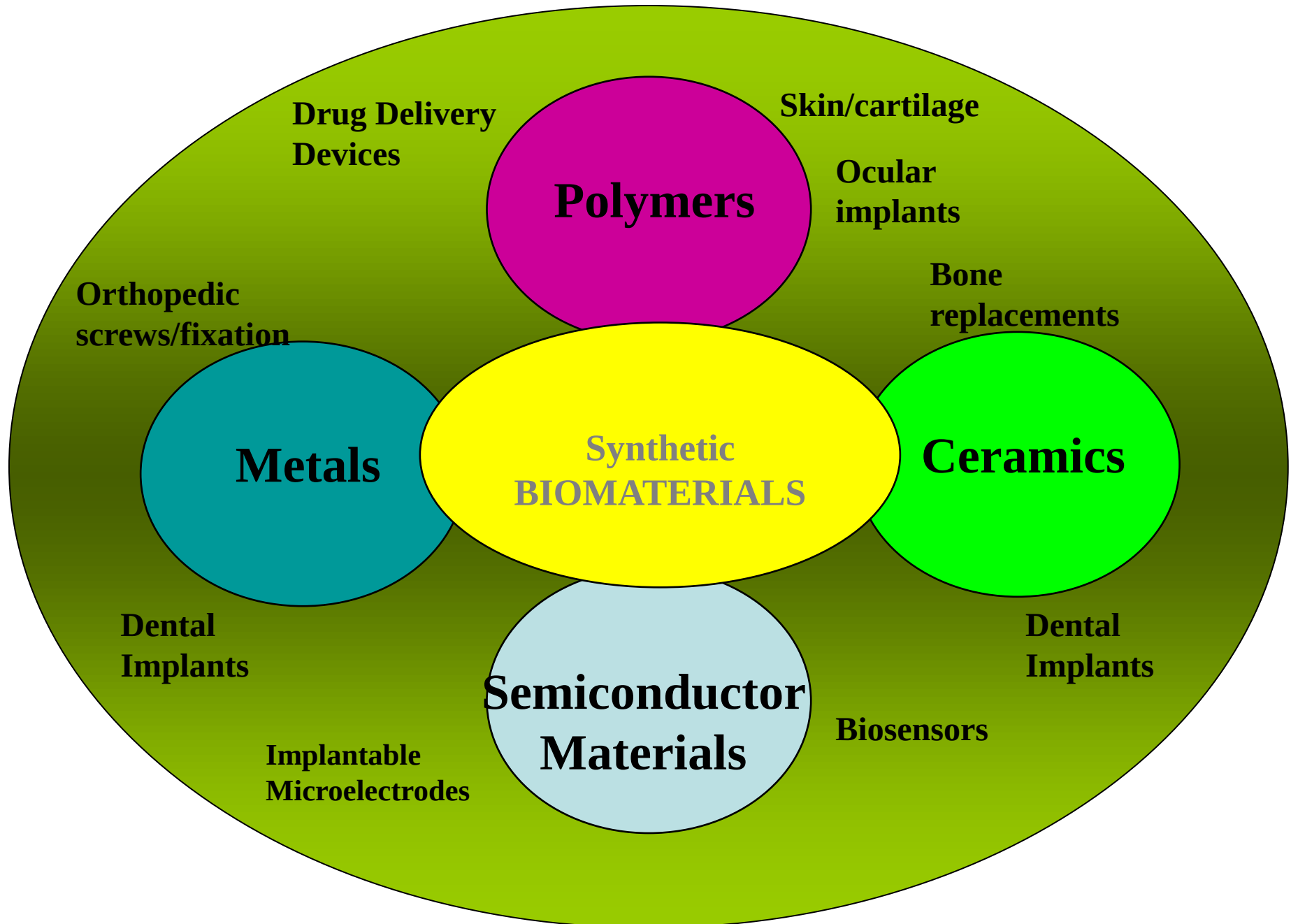


Chapter 13 Biological Properties (Bioceramics)



Bioceramics

Aims:

- To explore concepts and mechanisms related to Bioceramics;
- To examine Biological properties of ceramics;
- To introduce the use of ceramics as biomaterials;

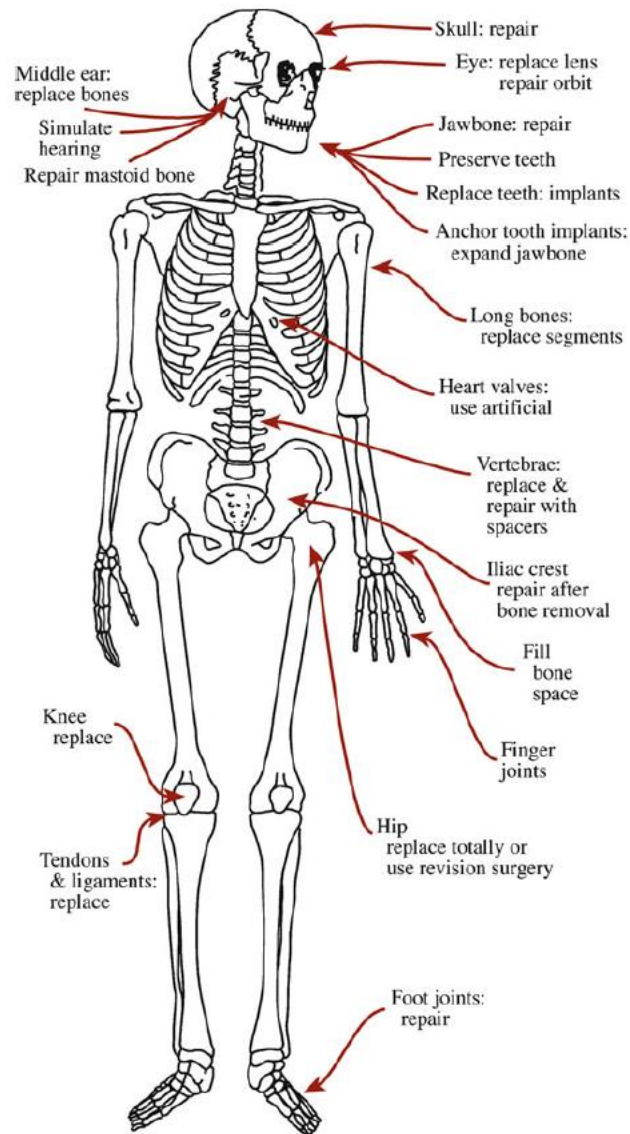


What are Bioceramics ?

Biomaterial: A non-viable material used in medical device intended to interact with biological systems.

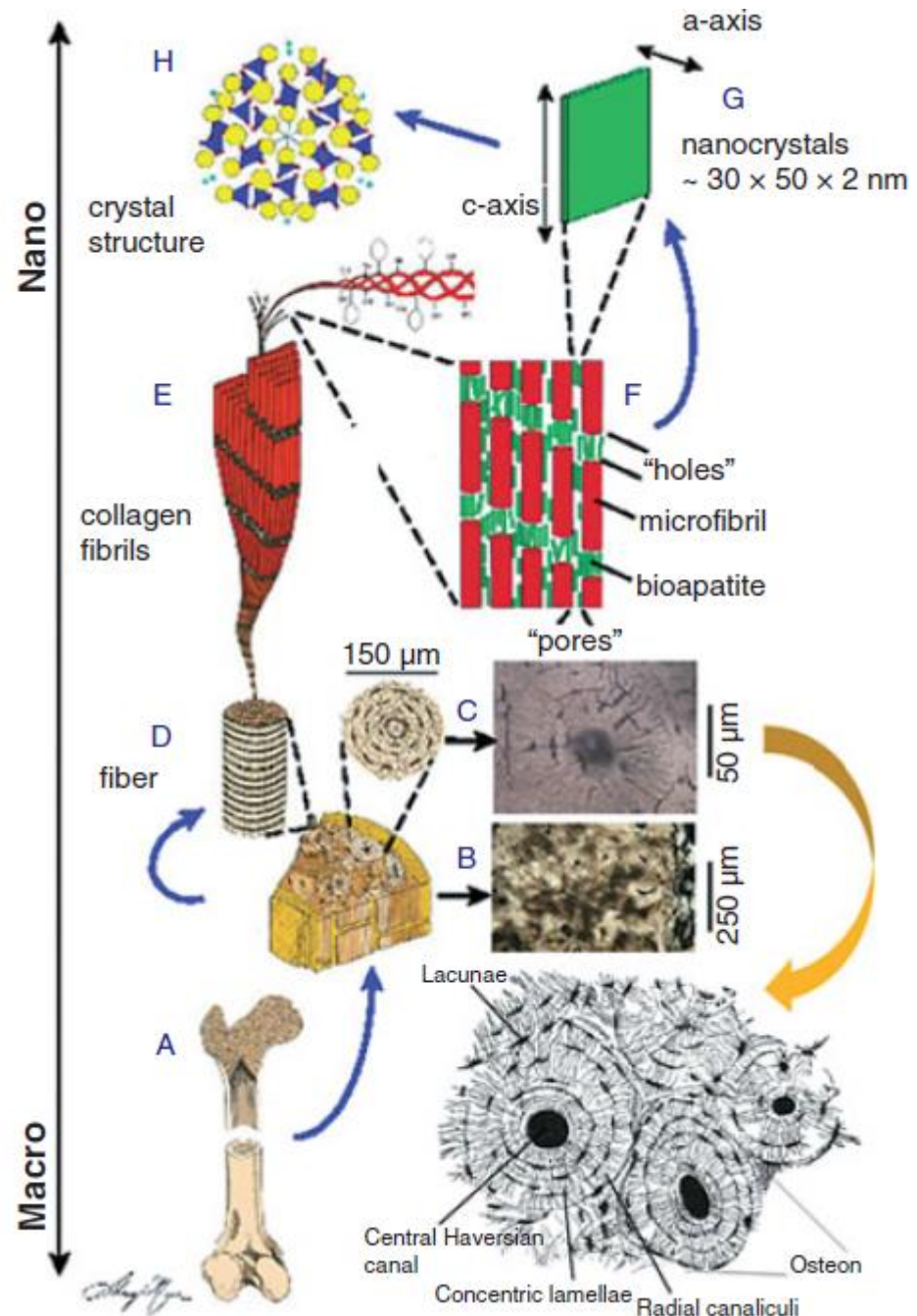
Bioceramic: A Ceramic used as a biomaterial. These ceramics are manufactured or processed to be suitable for use in or as a medical device that comes into intimate contact with proteins, cells, tissues, organs, and organ systems.

Why Bioceramics are important ?



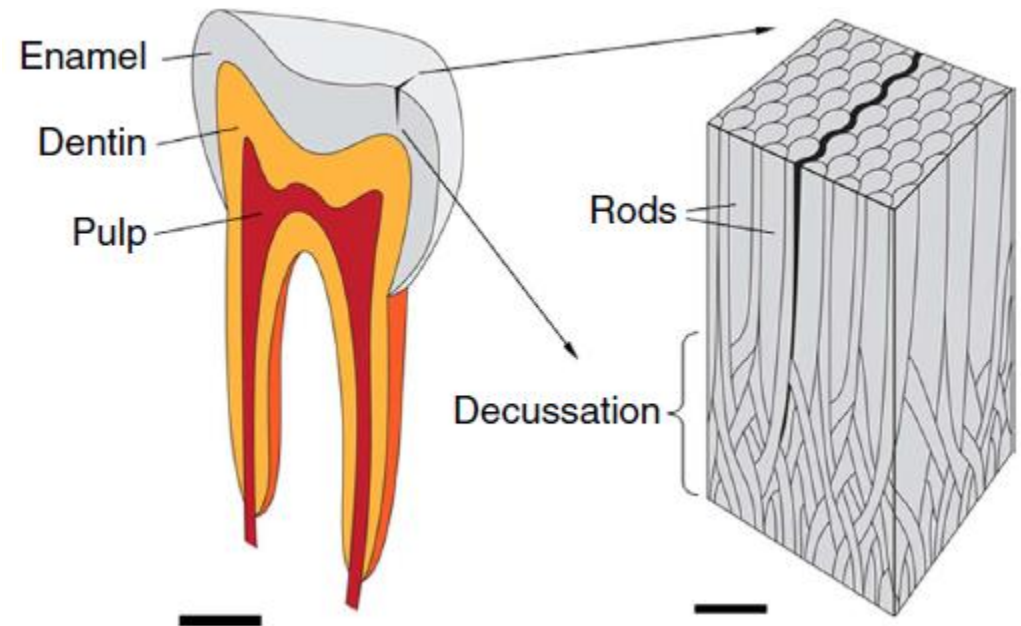
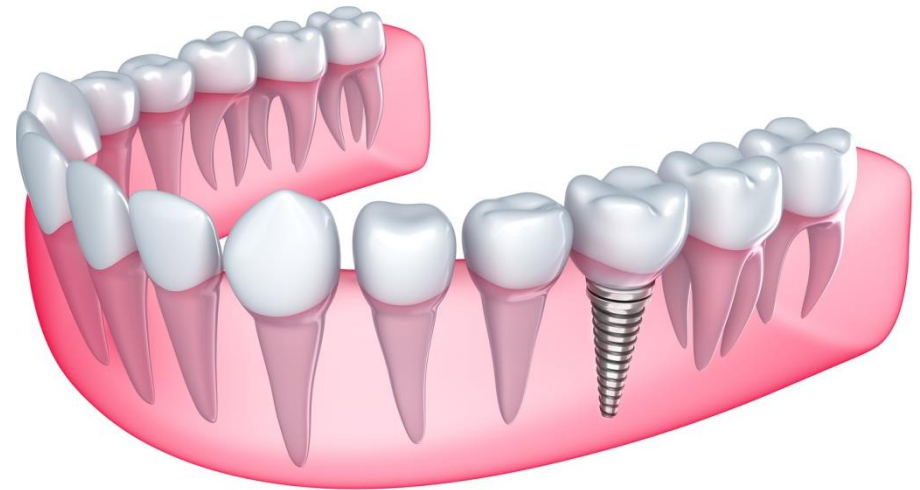
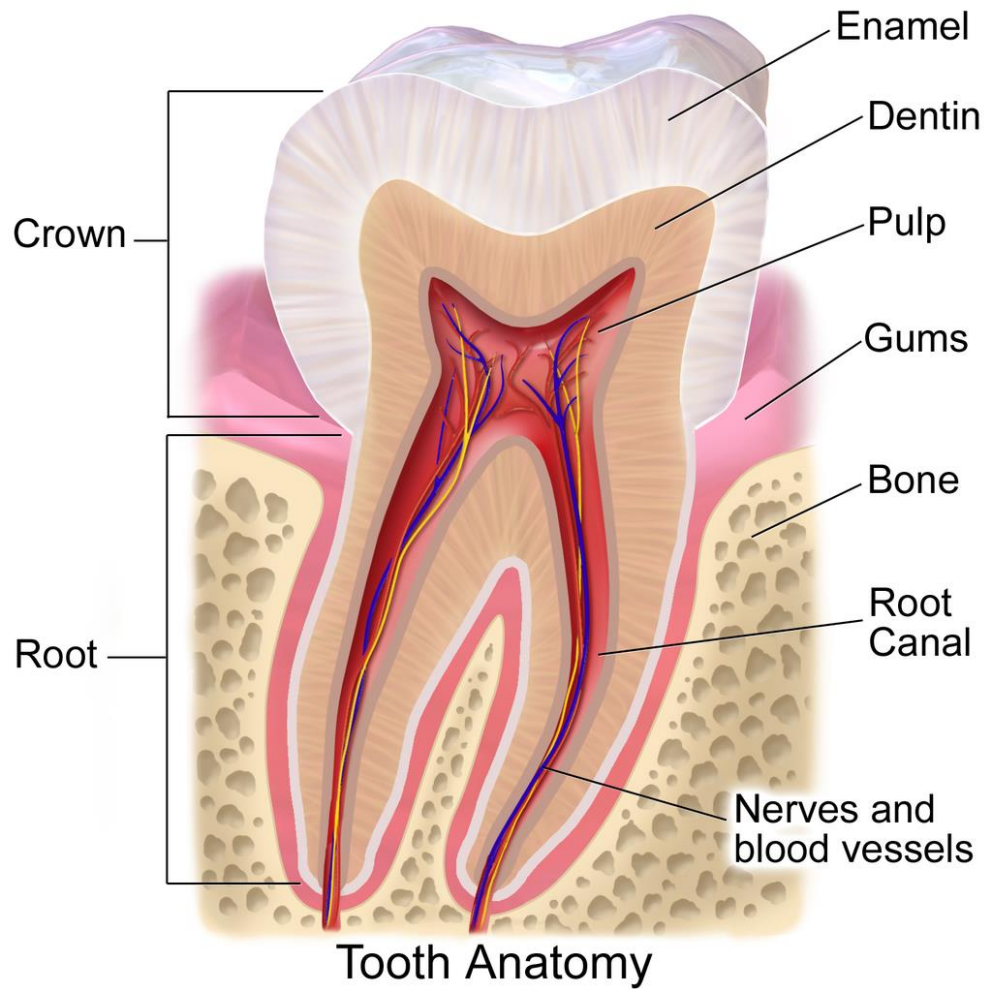
▮ The Bioceramic Man.

Ceramics are used from head to to as implants, repairs, and restorations. Courtesy of Professor Larry Hench, Imperial College, London.



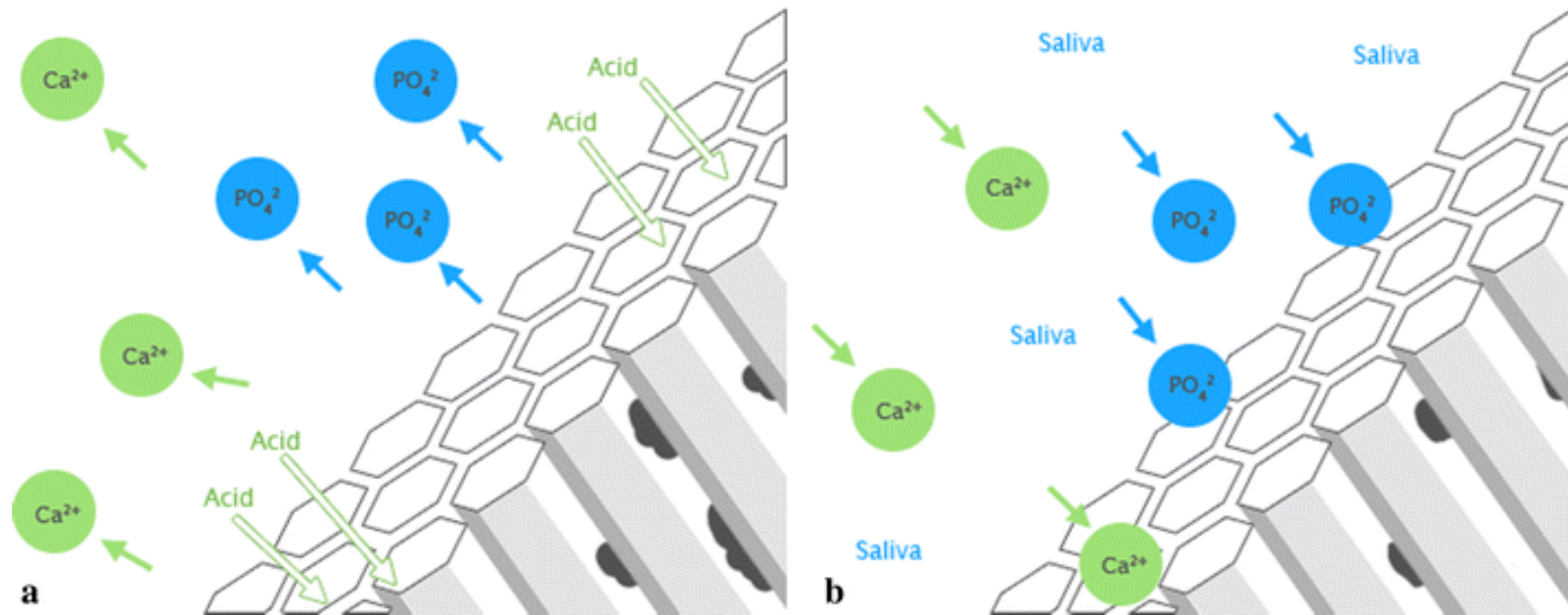
▮ Schematic of the hierarchical architecture of cortical bone.

Why Bioceramics are important ?



Biom mineralization

 Biom mineralization is the process by which living organisms produce minerals, often to harden or stiffen existing tissues. Such tissues are called mineralized tissues.



Schematic drawings of (a) demineralization and (b) remineralization processes of dental enamel

Biological Properties

Biocompatibility :

The ability of a material to perform with an appropriate host response in a specific application. Objective is to minimize inflammatory responses and toxic effects.

Bioactivity - Evolving concept:

- The characteristic that allows the material to form a develop a direct, adherent, and strong bond with living tissue (Hench, 1971)
- The ability of a material to stimulate healing and trick the tissue system into responding as if it were a natural tissue (Hench 2002).
- Advantages: Bone tissue – implant interface, enhanced healing response, extends implant life

Biodegradability:

- Breakdown of implant due to chemical or cellular actions
- If timed to rate of tissue healing transforms implant to scaffold for tissue regeneration
- Negates issues of stress shielding, implant loosening, long term stability

Biological Properties

Toxicity: the degree to which a substance can damage an organism. Toxicity can refer to the effect on a whole organism, such as an animal, bacterium, or plant, as well as the effect on a substructure of the organism, such as a cell (cytotoxicity) or an organ such as the liver (hepatotoxicity).

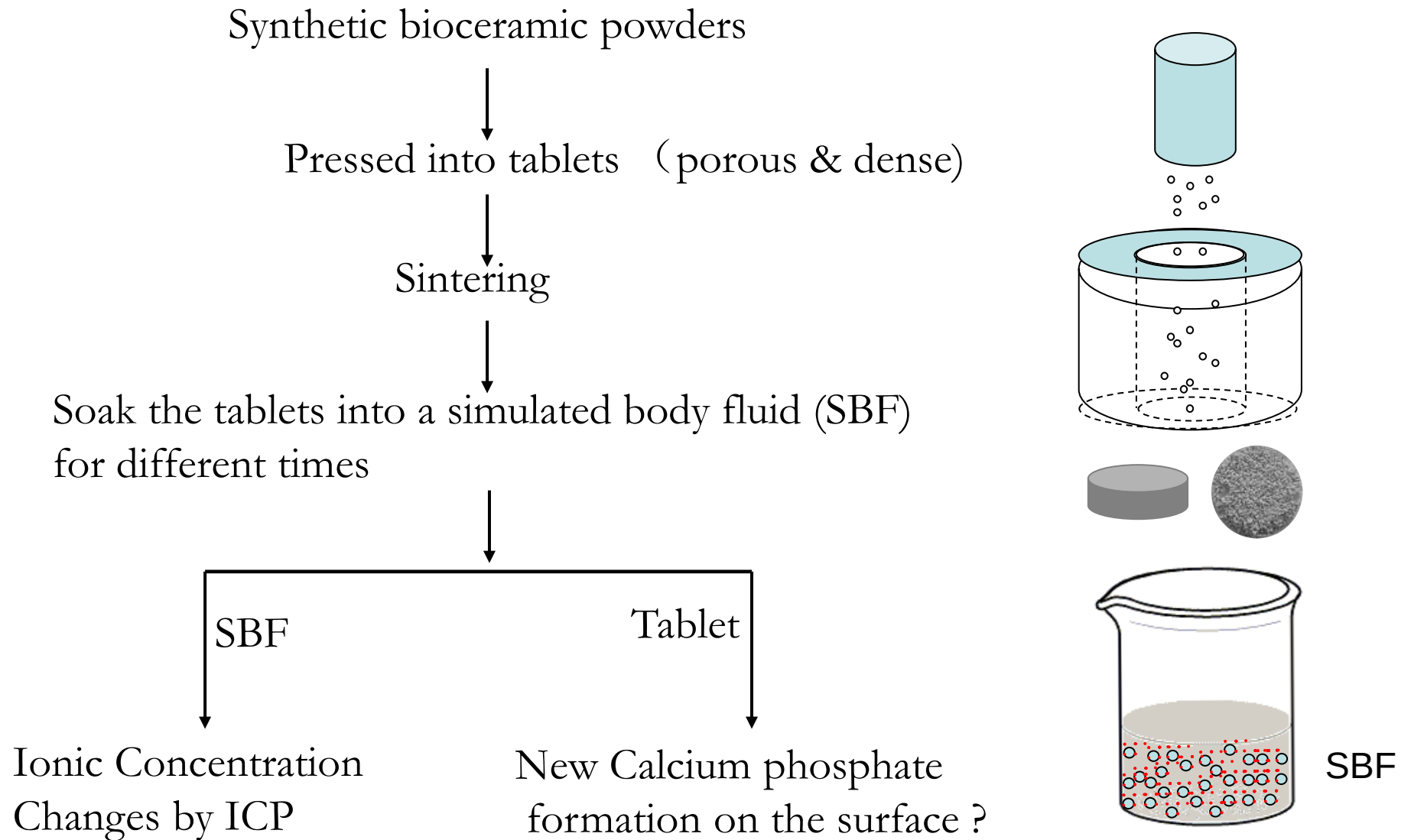
Osteoconductivity: the property of graft material in which it serves as a scaffold for new bone growth but does not induce bone growth de novo. This means that osteoconductive materials will only contribute to new bone growth in an area where there is already vital bone.

Osteoinductivity: the property of graft material in which it induces de novo bone growth with biomimetic substances, such as bone morphogenetic proteins. Such materials will contribute to new bone growth in an area where there is no vital bone, such as when implanted into muscle tissue.

Solution of simulated body fluid

<i>ion conc. (mM)</i>	Blood plasma	Revised SBF (r-SBF)
Na^+	142	142
K^+	5	5
Mg^{2+}	1.5	1.5
Ca^{2+}	2.5	2.5
Cl^-	103	103
HCO_3^-	27	27
HPO_4^{2-}	1	1
SO_4^{2-}	0.5	0.5
pH	7.2-7.4	7.4
<i>Organic components</i>	+	/
	Complex mixture	Salt solution

In vitro Bioactivity Evaluation



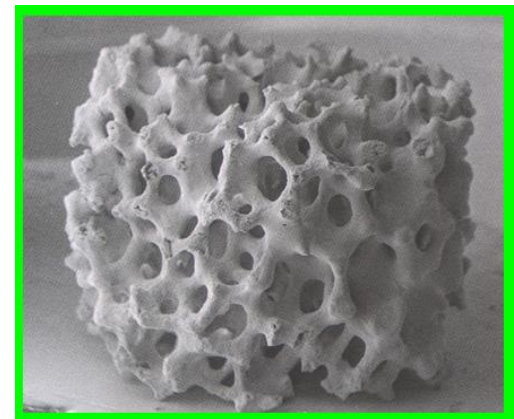
In vivo Osteo-inductivity Evaluation



Classification of Bioceramics

- Bioactive glass,
- Polycrystalline ceramics (alumina, calcium phosphates),
- Bioactive glass-ceramics,

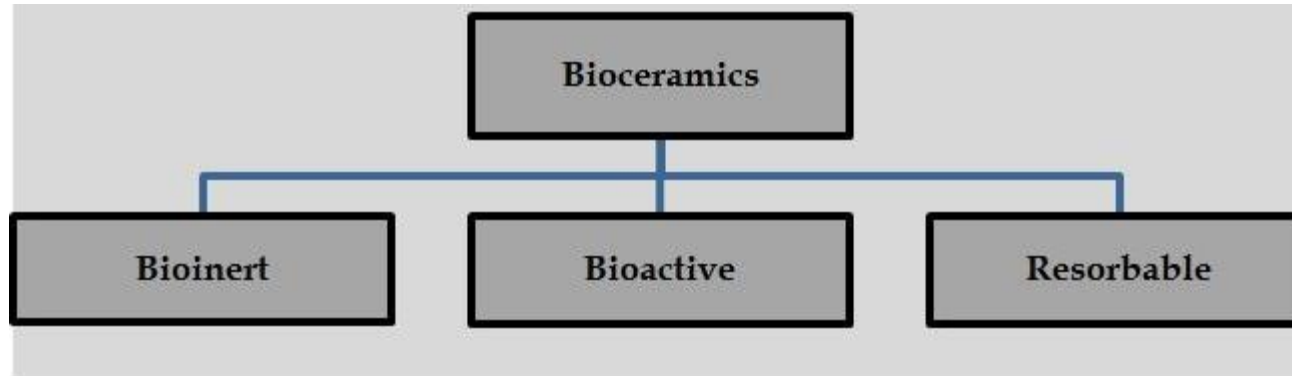
In the form of: Dense/Porous, granules/power, coatings, components of composites.



Classification based on tissue interaction

- Nearly Bioinert:
 - little response other than initial inflammatory response
- Bioactive:
 - The ability of a material to stimulate healing and trick the tissue system into responding as if it were a natural tissue (Hench 2002).
 - Advantages: integrated bone tissue – implant interface, enhanced healing response, extends implant life
- Biodegradable (resorbable):
 - Breakdown of implant due to chemical or cellular actions
 - If timed to rate of tissue healing, transforms implant to scaffold for tissue regeneration
 - Negates issues of stress shielding, implant loosening, long term stability

Classification based on Tissue Interaction



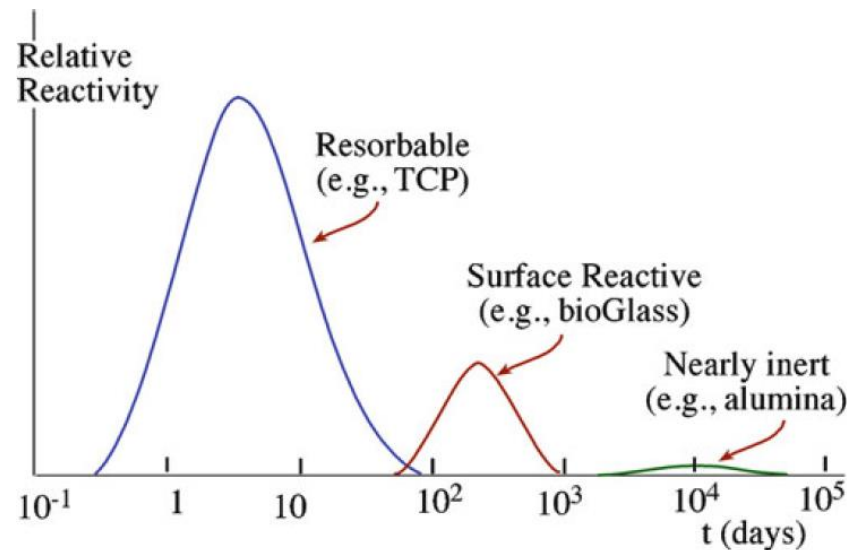
Bioinert Ceramics: result in little or no physiological reaction with human body and tend to exhibit inherently low levels of reactivity, such as alumina.

Bioactive ceramics: React in a positive way with local cells, directly attach by chemical bonds and have a substantial higher level of reactivity.

Resorbable Bioceramics: slowly and gradually replaced by surrounding tissue.

- Breakdown of implant due to chemical or cellular actions
- If timed to rate of tissue healing, transforms implant to scaffold for tissue regeneration;
- Negates issues of stress shielding, implant loosening, long term stability

Classification based on Tissue Interaction



Nearly inert bioceramics

Examples: Al_2O_3 , low temperature isotropic (LTI) carbon; ultra-LTI carbon; vitreous carbon; ZrO_2

Tissue attachment: mechanical

Bioactive ceramics

Examples: HA; bioactive glasses; bioactive glass-ceramics

Tissue attachment: interfacial bonding

Resorbable bioceramics

Examples: tricalcium phosphate (TCP); calcium sulfate; trisodium phosphate

Tissue attachment: replacement

Composites

Examples: HA/autogenous bone; surface-active glass ceramics/PMMA; surface-active glass/metal fibers; polylactic acid (PLA)/carbon fibers; PLA/HA; PLA/calcium/phosphorus-based glass fibers

Tissue attachment: depends on materials

Bioinert Ceramics: Alumina

History:

- since early seventies more than 2.5 million femoral heads implanted worldwide.
- alumina-on-alumina implants have been FDA monitored
- over 3000 implants have been successfully implemented since 1987

Smaller the grain size and porosity, higher the strength

- $E = 380 \text{ GPa}$ (stress shielding may be a problem)

Bioinertness

- Results in biocompatibility – low immune response
- Disadvantage:
 - Minimal bone ingrowth
 - Non-adherent fibrous membrane
 - Interfacial failure and loss of implant can occur



Properties and applications of Bionert Ceramics: Alumina

Properties:

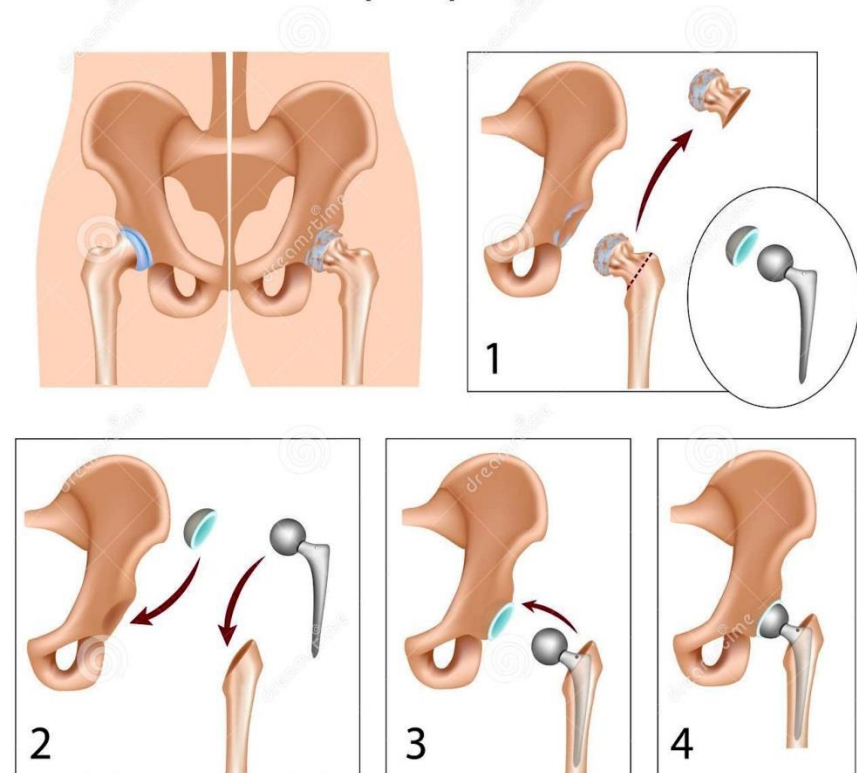
- scratch resistant
- low friction coefficient
- very low wear
- corrosion resistant
- highly wettable



Applications:

- hip implants, knee prostheses
- porous coatings for femoral stems
- porous bone spacers
- dental crowns and bridges

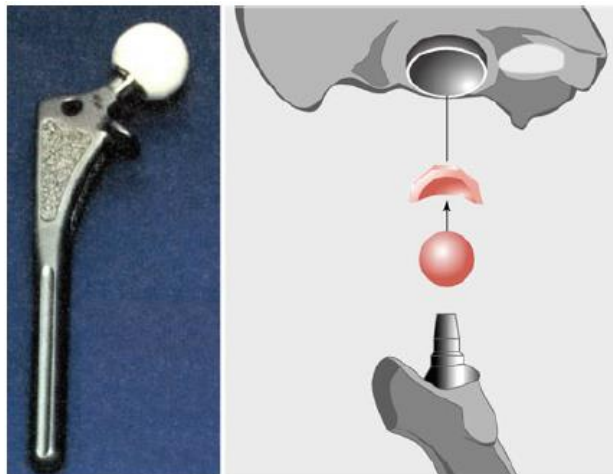
Total Hip Replacement



Bioinert Ceramics: Alumina (Al_2O_3)

Applications

- orthopaedics:
 - femoral head
 - bone screws and plates
 - porous coatings for femoral stems
 - porous spacers (specifically in revision surgery)
 - knee prosthesis
- dental: crowns and bridges



Bioinert Ceramics: Zirconia, ZrO_2

Fabrication:

- Obtained from the mineral zircon
- Addition of MgO , CaO , CeO , or Y_2O_3 stabilize tetragonal crystal structure (e.g. 97 mol% ZrO_2 and 3 mol% Y_2O_3)
- Usually hot-pressed or hot isostatically pressed



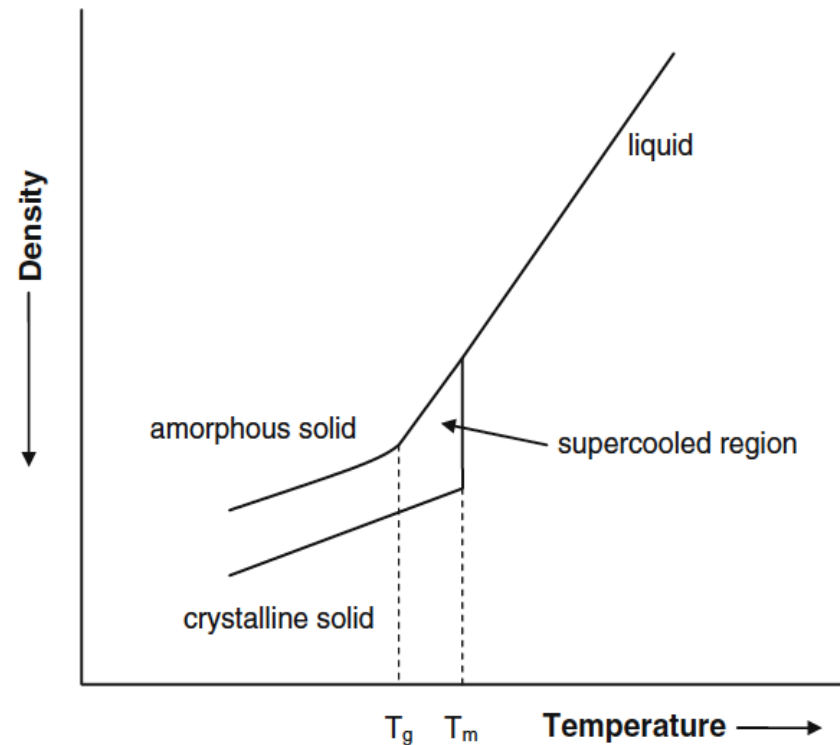
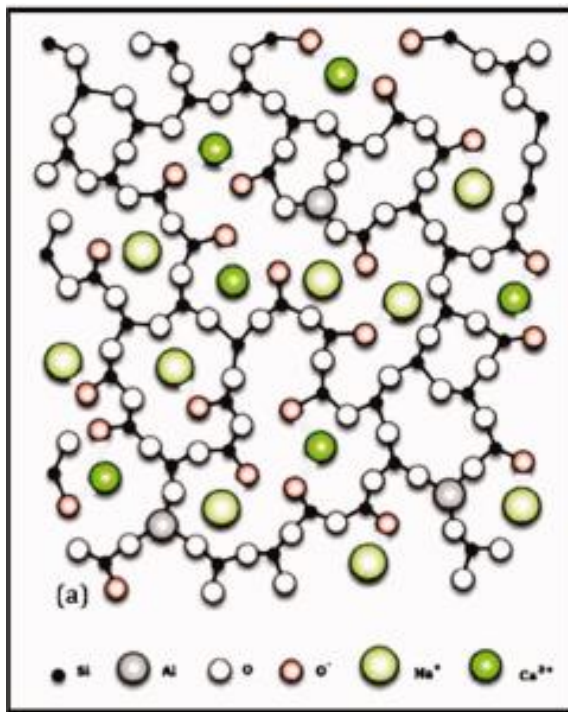
Applications:

- orthopedics: femoral head, artificial knee, bone screws and plates, favored over UHMWPE due to superior wear resistance
- dental: crowns and bridges



Bioactive Glass

- Glass:
 - an inorganic melt cooled to solid form without crystallization;
 - an amorphous solid;
- Glass-ceramic is a polycrystalline solid prepared by controlled crystallization of glass.



Desired Parameters for Bioactive Glass

- Biocompatibility of bioglasses is an indispensable property. They should be non-toxic and hence promote cell proliferation in addition to cell adhesion.
- For making scaffolds, almost every bioglass requires a thermal heat treatment, which leads to nucleation and growth of crystalline phases embedded in a matrix of glass. These crystallized phases must not induce cytotoxic effect or hamper any bioactive process inside the cell/tissue.
- When these glasses are in contact with SBF, there must be a formation of a hydroxyapatite layer.
- It should not exhibit any inflammatory response, demonstrate cytotoxicity or immunogenicity. Tissue scaffolds provide a temporary structure for cells to synthesize new tissue and they must exhibit neogenesis, however they must degrade into nontoxic products, which can be easily resorbed or be excreted by the body. Moreover, both the surface and the bulk material must be sterile.

Desired Parameters for Bioactive Glass

- The bioglass must possess required mechanical properties to withstand any kind of pressure or strain in order to prevent any structural failure during handling of the material and during the patient's normal routine activities. The bioglass scaffolds must exhibit mechanical properties that are comparable to those of the tissue to be replaced for better compatibility.
- For bone engineering, bioglass should possess controllable interconnected porosity to support vascularization so as to direct cells to grow into the required physical structure. A typical porosity of 90% along with a pore diameter of at least 100 μm is required for proper vascularization of the tissue.
- The architecture aspect of a bioglass scaffold should have a porous three-dimensional (3D) structure for cell proliferation, vascularization and diffusion of nutrients. This provides a regulated microenvironment for new tissue synthesis.
- For commercialization, the bioglass should be cost-effective while still maintaining the desired features.

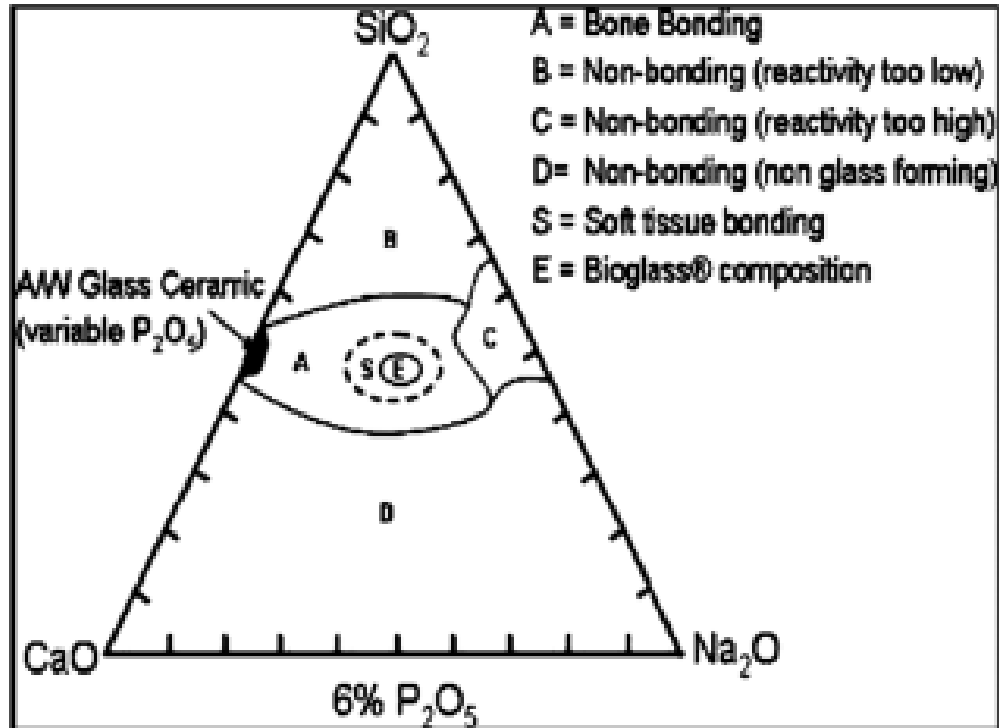
Bioactive Glass

- Glass ceramics were the first biomaterials to display bioactivity (bone system):
 - Capable of direct chemical bonding with the host tissue
 - Stimulatory effects on bone-building cells
 - Composition includes SiO_2 , CaO and Na_2O
 - Bioactivity depends on the relative amounts of SiO_2 , CaO and Na_2O
 - Cannot be used for load bearing applications
 - Ideal as bone cement filler and coating due to its biological activity

Compositions of various bioactive glasses.

Composition (wt.%)	45S5	13-93	6P53B	58S	70S30C	13-93B1	13-93B3	P ₅₀ C ₃₅ N ₁₅
Na ₂ O	24.5	6.0	10.3	0	0	5.8	5.5	9.3
K ₂ O	0	12.0	2.8	0	0	11.7	11.1	0
MgO	0	5.0	10.2	0	0	4.9	4.6	0
CaO	24.5	20.0	18.0	32.6	28.6	19.5	18.5	19.7
SiO ₂	45.0	53	52.7	58.2	71.4	34.4	0	0
P ₂ O ₅	6.0	4.0	6.0	9.2	0	3.8	3.7	71.0
B ₂ O ₃	0	0	0	0	0	19.9	56.6	0

Silicate Glass



Ternary compositional diagram given for 45% SiO₂-24.5% Na₂O-24.5% CaO-6%P₂O₅ glass by Hench for bone-bonding.

Bioglass® 45S5

A: Bonding within 30 days

B: Nonbonding, reactivity too low

C: Nonbonding, reactivity too high

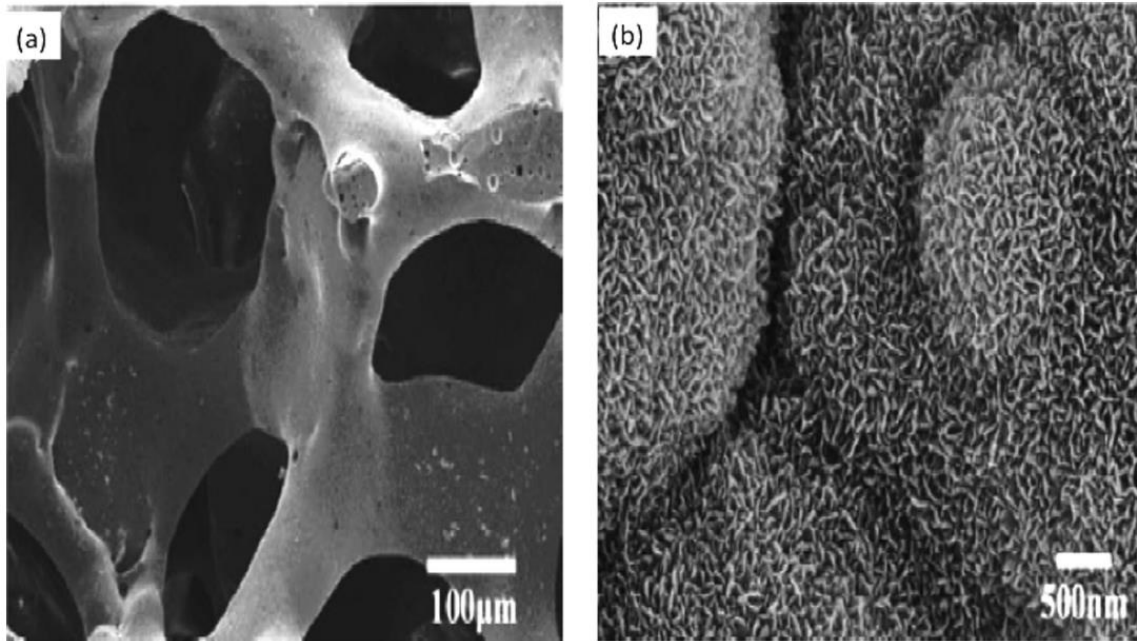
D: Bonding

📖 The 45S5 glass cannot be easily pulled into fibers because of its tendency to devitrify (crystallize). Thermal bonding (sintering) of 45S5 particles into anatomically relevant shapes requires temperatures of ~1000 °C and higher, which leads to devitrification to form a predominantly combeite crystalline phase (Na₂O · 2CaO · 3SiO₂).

📖 Larry Hench and colleagues at the University of Florida first developed these materials in the late 1960s and they have been further developed by his research team at the Imperial College London and other researchers worldwide.

Silicate Glass

13-93 Bioglass: Composition (wt.%) (53SiO_2 - $6\text{Na}_2\text{O}$ - $12\text{K}_2\text{O}$ - 5MgO - 20CaO - $4\text{P}_2\text{O}_5$)

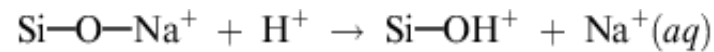


(a) 13-93 glass scaffold (b) formation of hydroxyapatite needles on the surface of 13-93 glass after 6 days in SBF.

📖 The glass can be pulled into fibers, and particles or short fibers have been sintered, without devitrification, to form porous scaffolds with anatomically relevant shapes.

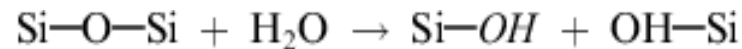
The mechanism of bioactivity and bone bonding of 45S5 glass

Stage 1: Rapid ion exchange reactions between the glass network modifiers (Na^+ and Ca^{2+}) with H^+ (or H_3O^+) ions from the solution, leads to hydrolysis of the silica groups and the creation of silanol ($\text{Si}-\text{OH}$) groups on the glass surface: e.g. equation(1)



The pH of the solution increases due to the consumption of H^+ ions.

Stage 2: The increase in pH (or OH^- concentration) leads to attack of the SiO_2 glass network, and the dissolution of silica, in the form of silicic acid, $\text{Si}(\text{OH})_4$, into the solution, and the continued formation of $\text{Si}-\text{OH}$ groups on the glass surface: equation(2)



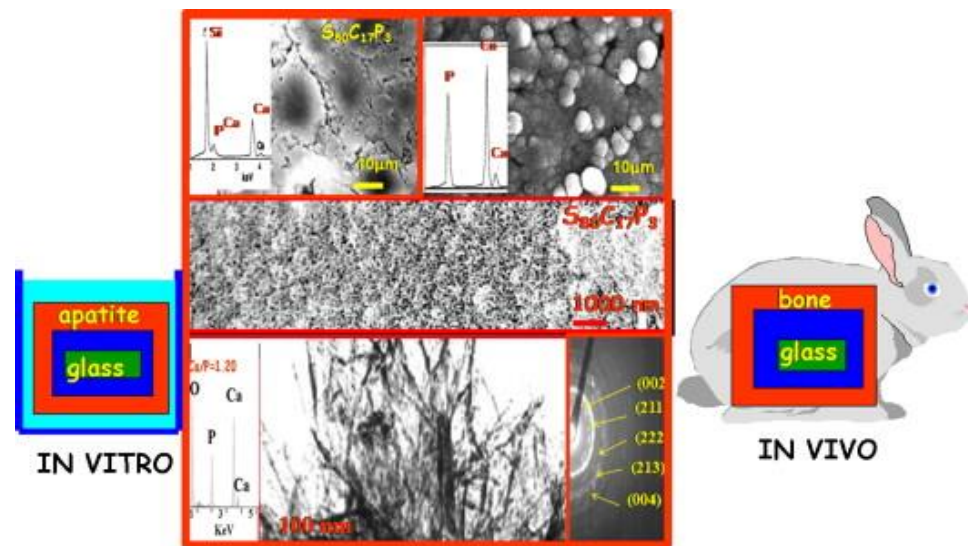
While the solubility of silica is low, the products of 45S5 glass and glass–ceramic dissolution in aqueous solutions have shown an increase in Si concentration, indicating that dissolution of silica is an important mechanism. However, other mechanisms could also contribute to the increase in Si concentration.

The mechanism of bioactivity and bone bonding of 45S5 glass

Stage 3: Condensation and polymerization of an amorphous SiO₂-rich layer (typically 1–2 μm thick) on the surface of the glass depleted in Na⁺ and Ca²⁺.

Stage 4: Further dissolution of the glass, coupled with migration of Ca^{2+} and $(\text{PO}_4)^{3-}$ ions from the glass through the SiO_2 -rich layer and from the solution, leading to the formation of an amorphous calcium phosphate (ACP) layer on the surface of the SiO_2 -rich layer.

Stage 5: The glass continues to dissolve, as the ACP layer incorporates $(\text{OH})^-$ and $(\text{CO}_3)^{2-}$ from the solution and crystallizes as an HCA layer.



Calcium Phosphate

- Powders
- Scaffolds
- Coatings for implants – metals, heart valves to inhibit clotting
- Self-Setting bone cement

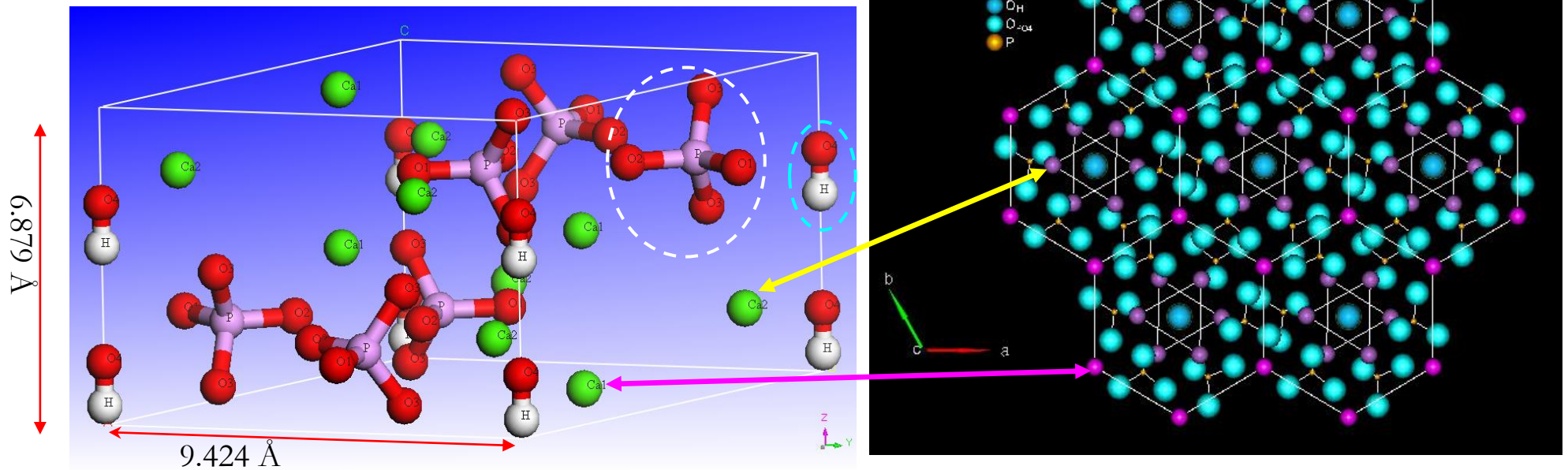
	Name	Acronym	Formula	Ca/P
1	monocalcium phosphate anhydrate	MCPA	$\text{Ca}(\text{H}_2\text{PO}_4)_2$	0.5
2(b)	dicalcium phosphate anhydrate	DCPA	CaHPO_4	1
3	octacalcium phosphate	OCP	$\text{Ca}_8(\text{HPO}_4)_2(\text{PO}_4)_4 \cdot 5\text{H}_2\text{O}$	1.33
4(a)	alpha tricalcium phosphate	TCP	$\text{Ca}_3(\text{PO}_4)_2$	1.5
4(b)	beta tricalcium phosphate	TCP	$\text{Ca}_3(\text{PO}_4)_2$	1.5
5	amorphous calcium phosphate	ACP	$\text{Ca}_x(\text{PO}_4)_y \cdot n\text{H}_2\text{O}$	1.1-1.5
6(a)	hydroxyapatite	HA	$\text{Ca}_5(\text{PO}_4)_3\text{OH}$	1.67
6(b)	calcium deficient hydroxyapatite	cd-HA	$\text{Ca}_9(\text{HPO}_4)(\text{PO}_4)_2\text{OH}$	1.5
7	tetra calcium phosphate	TetCP	$\text{Ca}_4(\text{PO}_4)_2\text{O}$	2

Calcium-phosphate ceramics ; their structure is the form of multiple oxides of calcium and phosphate atoms

Hydroxyapatite (HA)

☯ Chemical Formula: $\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2$, $\text{P6}_3/\text{m}$

☯ Crystal structure of HA



☯ Basic mineral component of bone, enamel, and dentin

☯ Bones or teeth substitute materials

☯ Template for Substitution: $\text{M}_{10}(\text{PO}_4)_6\text{X}_2$, $\text{P6}_3/\text{m}$

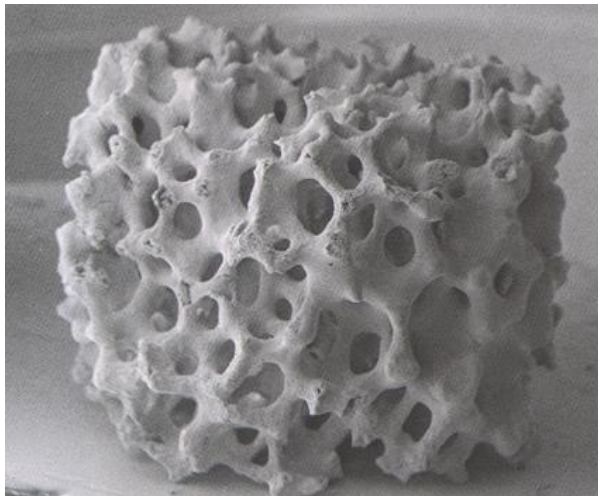
Biological Apatites

📌 Comparative composition of Human Enamel, Bone and HA Ceramic

Constituents (wt%)	Enamel	Bone	HA
Calcium, Ca^{2+}	36.0	24.5	39.6
Phosphorus, P	17.7	11.5	18.5
(Ca/P) molar	1.62	1.65	1.67
Sodium, Na^+	0.5	0.7	tr
Potassium, K^+	0.08	0.03	tr
Magnesium, Mg^{2+}	0.44	0.55	tr
Carbonate CO_3^{2-}	3.2	5.8	-
Fluoride, F^-	0.01	0.02	-
Chloride, Cl^-	0.30	0.10	-
Ash (total inorganic)	97.0	65.0	100
Absorbed H_2O	1.0	25.0	-
Trace elements: Sr^{2+} , Zn^{2+} , Ba^{2+} , Fe^{3+} , etc	1.5	9.7	-

Biodegradable Ceramics: Calcium Phosphate

- Structure resembles bone mineral; thus used for bone replacement
- Coating of metal implants to promote bone ingrowth
- Different forms exist depending on Ca/P ratio, presence of water, impurities and temperature
- 7 different forms of PO_4 based calcium phosphates exist - depend on Ca/P ratio, presence of water, pH, impurities and temperature

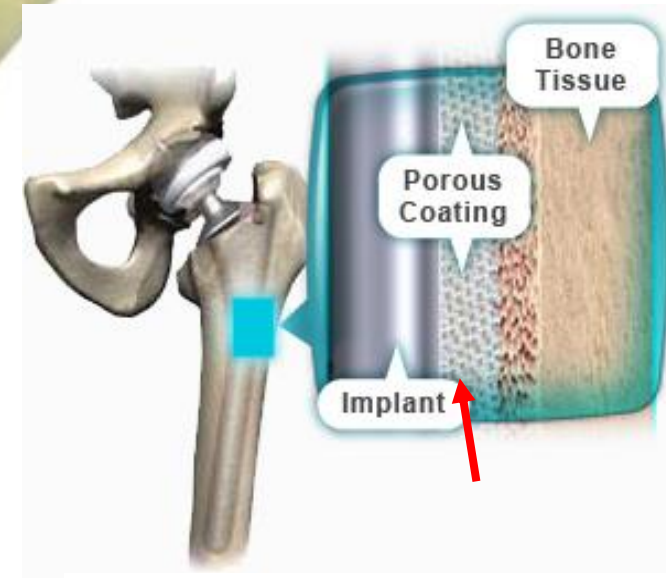


Biodegradation of Calcium Phosphate

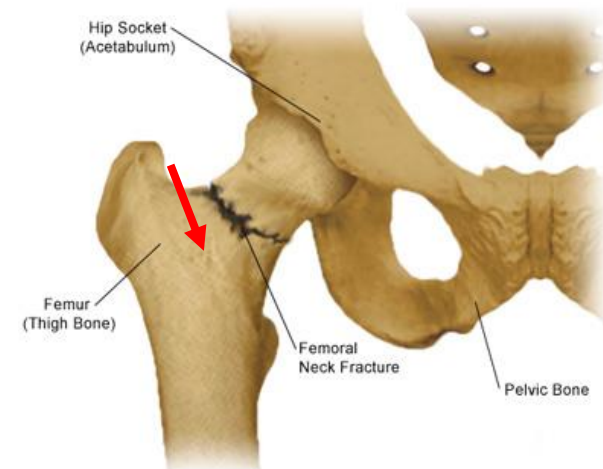
- Degradation caused by:
 - physiologic dissolution (depends on environment pH, type of calcium phosphate)
 - physical disintegration (grain boundary attack)
 - biological factors (phagocytosis etc.)

Potential Applications of Calcium Phosphates

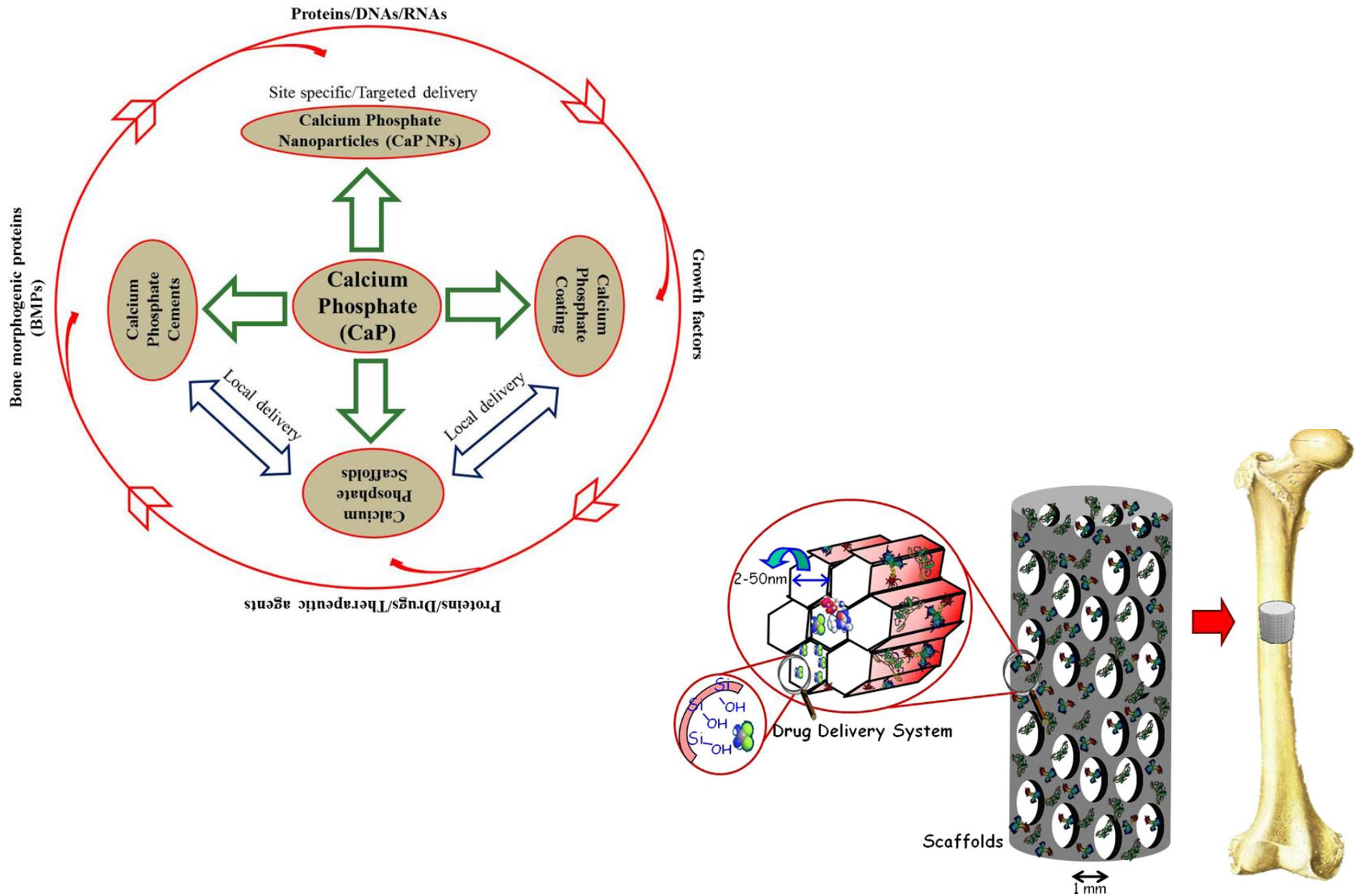
- Bone substitutes;
- As coatings for metallic prostheses;
- Repair of the fracture caused by osteoporosis;
- Other non-medical applications (catalysts, gas sensors, et al)



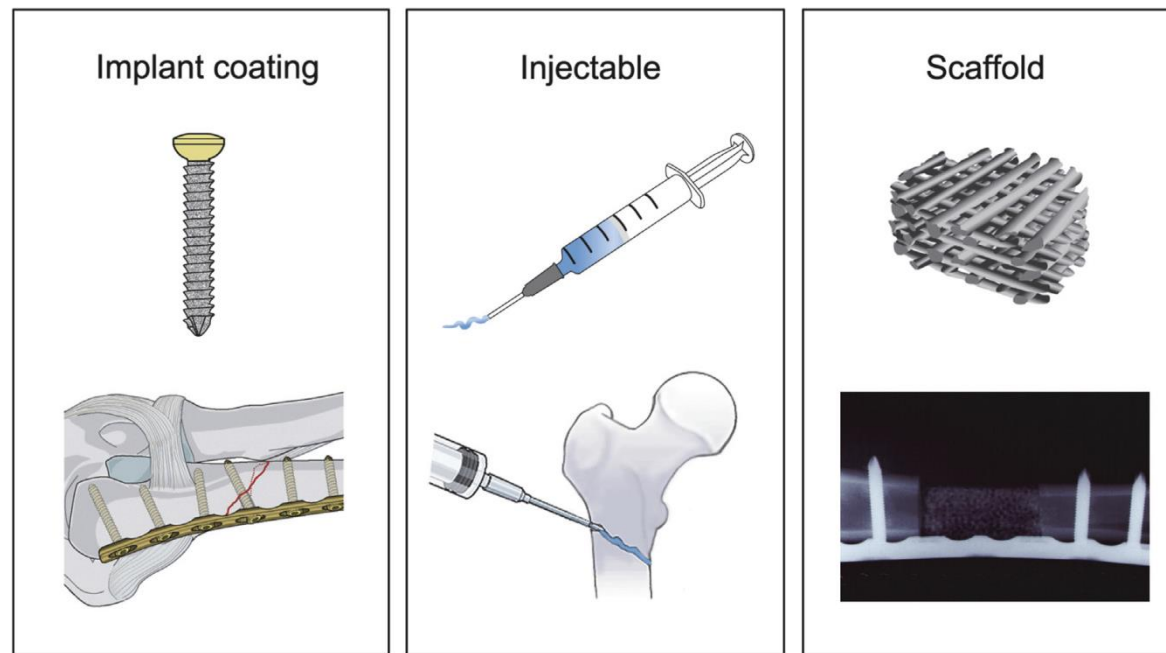
Femoral Neck Fracture



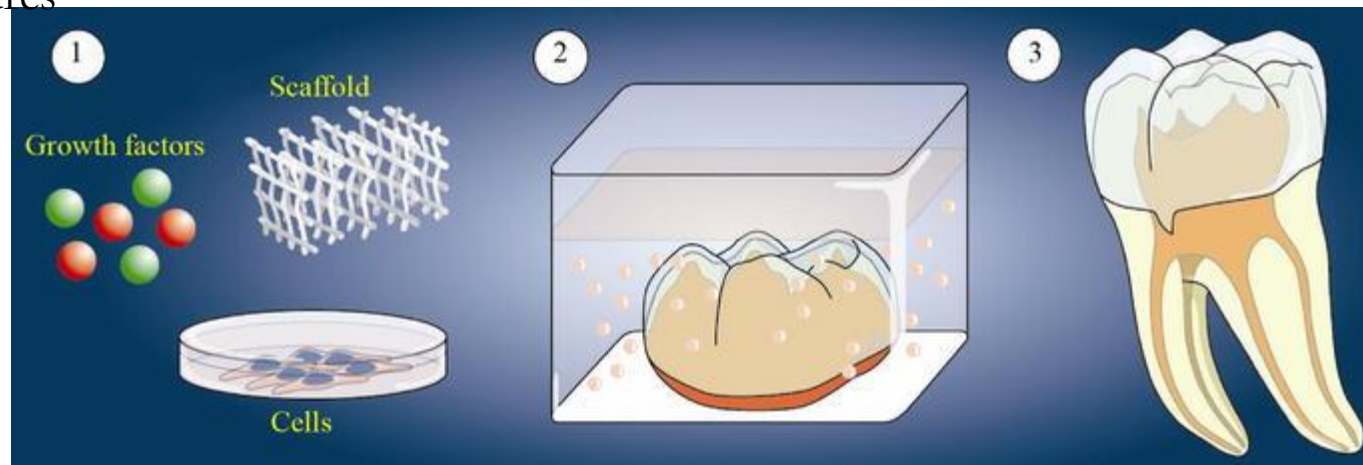
Calcium Phosphates for local drug delivery



Calcium Phosphates for local drug delivery



☐ Schematics of potential local drug delivery strategies for osteoporotic fractures

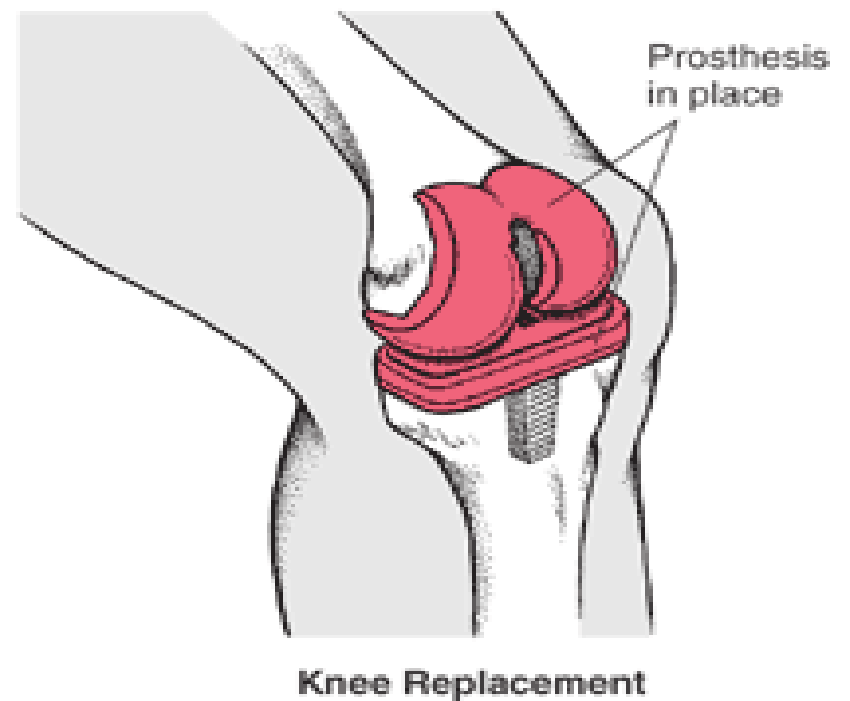
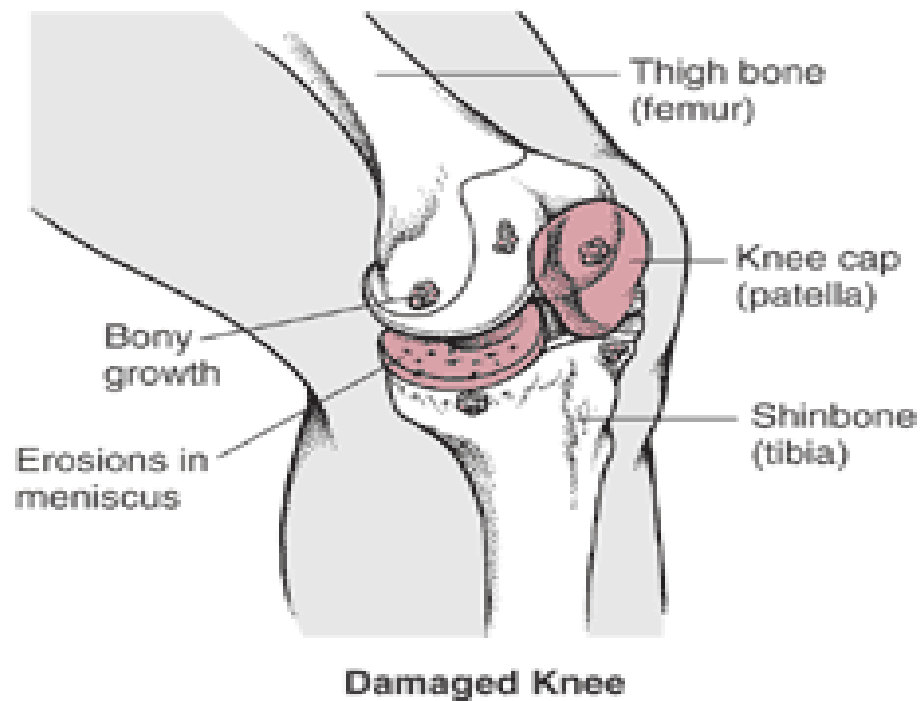


☐ A schematic diagram of entire tooth regeneration from the proper combination of growth factors and cells (stem cells or progenitor cells) seeded on a calcium orthophosphate scaffold.

Interaction of bioceramics with tissues

- All materials placed on live tissue ,takes response from tissue

TISSUE - IMPLANT



Advantage and disadvantage of Bioceramics

Advantage:

Biological properties: Biocompatible, biodegradable, non-toxic, bioactive, corrosion resistant.....

Disadvantage:

Low strength, Brittle, uncertainties to durability in different strokes and pressures....

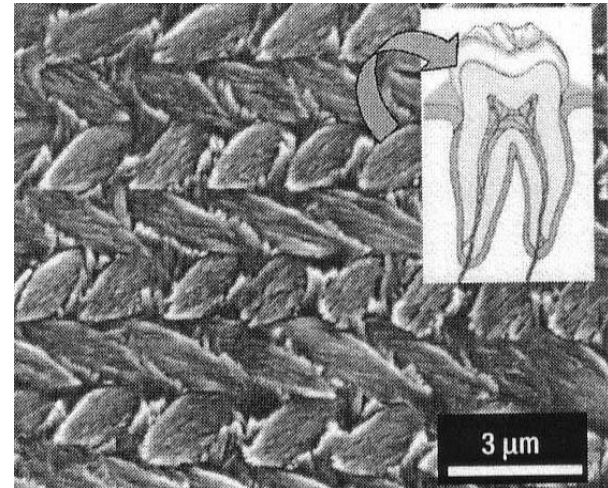
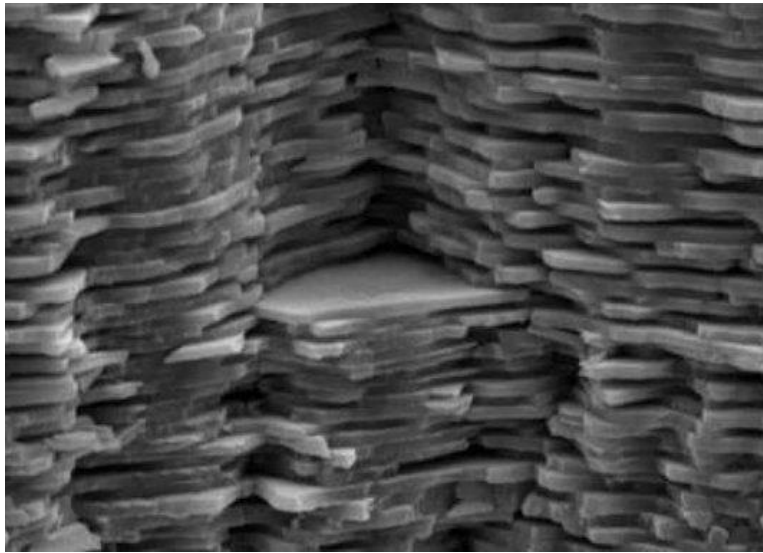
Mechanical Properties

<i>Material</i>	<i>Density (g/cm³)</i>	<i>UTS (MPa)</i>	<i>Compressive strength (MPa)</i>	<i>E (GPa)</i>	<i>K_{1c} (MPa· m^{1/2})</i>	<i>Hardness (Knoop)</i>	<i>α (ppm/°C)</i>	<i>Fracture surface energy (J/m²)</i>	<i>Poisson's ratio</i>	<i>k (Wm⁻¹K⁻¹)</i>
HA	3.1	40–300	300–900	80–120	0.6–1.0	400–4,500	11	2.3–20	0.28	
TCP	3.14	40–120	450–650	90–120	1.20		14–15	6.3–8.1		
Bioglasses	1.8–2.9	20–350	800–1,200	40–140	~2	4,000–5,000	0–14	14–50	0.21–0.24	1.5–3.6
A-W glass ceramic	3.07	215	1,080	118	2					
SiO ₂ glass	2.2	70–120		~70	0.7–0.8	7,000–7,500	0.6	3.5–4.6	0.17	1.5
Al ₂ O ₃	3.85–3.99	270–500	3,000–5,000	380–410	3–6	15,000–20,000	6–9	7.6–30	0.27	30
PSZ	5.6–5.89	500–650	1,850	195–210	5–8	~17,000	9.8	160–350	0.27	4.11
Si ₃ N ₄	3.18	600–850	500–2,500	300–320	3.5–8.0	~22,000	3.2	20–100	0.27	10–25
SiC	3.10–3.21	250–600	~650	350–450	3–6	~27,000	4.3–5.5	22–40	0.24	100–150
Graphite	1.5–2.25	5.6–25	35–80	3.5–12	1.9–3.5		1–3	~500	0.3	120–180
LTI-ULTI	1.5–2.2	200–700	330–360	25–40			1–10		0.3	2.5–420
Carbon fiber	1.5–1.8	400–5,000	330–360	200–700						
Glassy carbon	1.4–1.6	150–250	~690	25–40		8,200	2.2–3.2			
PE	0.9–1.0	0.5–65		0.1–1.0	0.4–4.0	170	11–22	500–8,000	0.4	0.3–0.5
PMMA	1.2	60–70	~80	3.5	1.5	160	5–8.1	300–400		0.20
Ti	4.52	345	250–600	117	60	1,800–2,600	8.7–10.1	~15,000	0.31	
Ti/Al/V alloys	4.4	780–1,050	450–1,850	110	40–70	3,200–3,600	8.7–9.8	~10,000	0.34	
Ti/Al/Nb/Ta alloys	4.4–4.8	840–1,010		105	50–80			~17,000	0.32	
Vitallium-stellite alloys (Co-Cr-Mn)	7.8–8.2	400–1,030	480–600	230	120–160	3,000	15.6–17.0	~5,000	0.30	
Low C steel Fe-Cr-Ni alloys	7.8–8.2	540–4,000	1,000–4,000	200	55–95	1,200–9,000	16.0–19.0	~50,000	0.20–0.33	46

How to Improve the Mechanical Properties of Bioceramics ?

Inspired by Nature

- Natural hard tissues are “ceramic”-**polymer** composites:
 - Bones, Teeth



Nacre: Aragonite nanobricks provide structure and stability which are separated by layers of elastic biopolymer that give the nacre flexibility and durability.

Overview: Medical Miracles Yet to Come

 Ceramics have had a profound impact on medicine just during the present generation. Such uses for ceramics as in the glass microsphere treatment of cancer, laser surgery, endoscopic diagnosis and surgery, and high-resolution CT scans not only seem magical but also are truly miraculous. You can look forward to many more ceramic medical breakthroughs during the next generation.

 Radiation and drug delivery systems will be refined and new ones discovered, joint implants will continue to be refined, and prosthetics will respond more and more like the actual limb. And someday, maybe ceramic sensors and advanced electronics will continually monitor our health and provide feedback by way of a miniature telemetry transmitter to a hospital computer that can alert a doctor if dangerous symptoms arise.